

SURGICAL MOTION TRACKING SYSTEM

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Current Progress Update

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Low-cost motion capture real time skill feedback for surgical training

PROJECT DIRECTION

Goal:

Create a small scaled motion tracking system that can capture instrument movement and return useful training feedback without relying on expensive commercial systems.

Our current scope:

- Use a camera, compact motion simulator, and local processing hardware.
- Measure position, velocity, acceleration, smoothness, and task timing.

SYSTEM ARCHITECTURE

This 4 step process turns camera video into clear training feedback.

Capturing motion

A camera records the instrument as it moves. The system can use small markers or MediaPipe to identify the instrument.

Tracking the instrument

The software follows the instrument in each video frames. Filtering reduces noise and pose estimation finds its position.

Measure performance

The system checks how quickly and smoothly the task is completed and checks whether the instrument follows the intended path.

Give feedback

The results appear on a monitor or tablet. The trainee can review the metrics and see alerts for errors.

The system records movement, analyzes it, evaluates performance, and returns feedback to the trainee.

LOW COST DEPLOYMENT OPTIONS

Current component research identifies practical options for a compact training setup.

Component	Low cost option	Purpose / note
Camera	Consumer webcam, GoPro, or Raspberry Pi camera	Maintain roughly 30-60 FPS for responsive tracking
Motion simulator	Miniaturized Stewart platform (\$250)	Fits on a desktop and can simulate organ motion
Markers (Unlikely)	3D printed ABS markers or colored tape	Low intrusion and reusable for marker based tests
AI processing	Laptop GPU or edge device, such as Jetson Nano / RTX 3060	Supports real time inference
Software	OpenCV + MediaPipe + PyTorch	Full processing pipeline without operation system level restrictions

IMPLEMENTATION PROGRESS

The design has been translated into a staged build and test plan

1. Set up workspace: Mount camera and position a small surgical instrument phantom or low cost board.
2. Enable detection: Attach markers or select the markerless approach based on precision and budget.
3. Calibrate cameras: Map camera coordinates to a workspace reference frame.
4. Process motion data: Filter noise and estimate trajectories and kinematic parameters.
5. Deliver feedback: Display metrics and issue optional alerts for errors or deviations.

PERFORMANCE AND DESIGN CONSIDERATIONS

Evaluation will balance responsiveness with accuracy, safety, and scalability

Latency	Processing below 100-200 ms supports near real time training feedback.
Accuracy	Markerless tracking may lose millimeter level precision but can still support skill assessment.
Safety	The setup should stay nonintrusive and avoid interfering with surgical instruments.
Scalability	Additional cameras or sensors can increase precision without redesigning the core architecture.

EVIDENCE BASE AND NEXT STEPS

Existing tools support the feasibility of the proposed low cost approach.

Supporting examples:

- SPRK miniature Stewart platforms support motion simulation for surgical training.
- MediaPipe Hand Landmarker task model supports real time hand pose estimation with mobile deployment compatibility.
- KCF and ResNet style methods offer practical options for visual motion tracking.
- Trinocular MoCap systems show that multiple instruments can be captured at the same time.

Next Steps:

- Train a YOLO Pose model (YOLO26n-pose) to detect forceps.
- Train a model with one of our datasets to test.
- Match data collected by camera to the datasets.

THANK YOU

